

Nima Ghourchian

Karaj, Iran

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Education

Islamic Azad University – Karaj Branch

Sep 2021 – Feb 2026

Bachelor of Mechanical Engineering | GPA: 16.90/20

Relevant coursework: Statics: 18/20, Dynamics: 17/20, Mechanical vibrations: 19.5/20, Automatic Control: 19/20, Machine Dynamics: 20/20, Engineering Mathematics: 18/20, Differential Equations: 20/20, Numerical Analysis: 19/20, Fundamentals of Electrical Engineering 1: 20/20

Selected Projects

National CubeSat Design Competition – Qsat

Jan 2024 – Jul 2026

Attitude Determination & Control member | Sharif University Team (Cubisa)

- Advanced to the final four among 52 teams nationwide to receive funding to build an engineering prototype and ultimately won third place.
- Completed a 180-hour CubeSat Design workshop covering topics such as ADCS, orbital mechanics, and mission design.
- Researched and implemented B-dot detumbling and quaternion-based PD feedback controller for nadir pointing in Simulink.
- Conducted Software-in-the-Loop (SIL) and Processor-in-the-Loop (PIL) testing to verify control algorithms and embedded implementation.

Vision-Based GPS-denied Navigation for UAVs

Jun 2025 – Jan 2026

B.Sc. Final Project

- Built a localization pipeline to match simulated UAV imagery with georeferenced satellite images using ResNet and LiteSAM to generate latitude and longitude estimates.
- Designed and implemented a Kalman filter that fuses simulated IMU, barometer and vision-based pose measurements.

Professional Experience

SpaceOmid

Sep 2025 – Apr 2026

Flight Control & Simulation Engineer – Omidspac Team

- Collaborated with a team of engineers to test and debug a spraying module prototype.
- Extended the existing manual altitude-control system by integrating an obstacle-responsive module that enabled automatic altitude adjustment during flight.

SpaceOmid

Jul 2025 – Sep 2025

Intern – Omidspac Team

- Participated in and contributed to weekly technical meetings.
- Assisted in sensor calibration, PID tuning of flight controllers and regular test flights.

Technical Skills

Programming Languages: MATLAB/Simulink, Python, C/C++

Frameworks & Tools: ROS 2, ArduPilot, PX4, AirSim, Unreal Engine

Software: SolidWorks, STK, Ansys

Languages

Persian: Native

English: IELTS Academic 7.0 (C1 Level)

